C-F-160a
SEPTEMBER 23, 1959
SUPERSEDING
Int. Fed. Spec. C-F-00160(GSA-FSS)
August 5, 1959 and
Fed. Spec. C-F-151a (In Part)
July 24, 1940
(See 6.4)

FEDERAL SPECIFICATION

FEATHERS, WATERFOWL AND DOWN, WATERFOWL

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers feathers and down from domesticated waterfowl (see 6.1).

1.2 Classification.

1.2.1 *Types, classes, and grades.*—Waterfowl feathers and down, shall be of the following types, classes, and grades as specified (see 6.2).

Type I.—Feathers.

Class 1.—Goose feathers.

Grade A.

Grade B.

Class 2.—Duck feathers.

Grade A.

Grade B.

Type II.—Down.

Class 1.—Goose down.

Grade A.

Grade B.

Class 2.—Duck down.

Grade A.

Grade B.

Type III.—Feathers and down mixture.

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

V-T-276—Thread, Cotton.

QQ-S-781—Strapping, Flat, Steel.

RR-S-366—Sieves, Standard for Testing Purposes.

CCC-C-429—Cloth, Cotton, Osnaburg.

CCC-T-191—Textile Test Methods.

PPP-B-636—Boxes, Fiber.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

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Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).
Fed. Std. No. 751—Stitches; Seams; and Stitching.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specifications:

JAN–C–485—Cloth, Cotton, Sheeting.
MIL–C–3985—Compound, Mothproofing, Emulsifiable Concentrate (for Textiles and Feathers).

Military Standards:

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.
MIL–STD–129 — Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Laboratory report approval.—Unless otherwise specified (see 6.2), prior to or upon submission of bids to the contracting officer, the prospective contractor shall furnish the contracting officer a certified copy of a recent laboratory report from any of the following:

(*a*) An independent or commercial laboratory report.
(*b*) The prospective contractor's own laboratory report.
(*c*) A Government laboratory report from a recent contract.

The laboratory report shall contain test data which demonstrates that the feathers, down, or feather and down mixture, of the type which the contractor proposes to deliver, have been previously tested and found to comply with all the requirements of this specification.

3.2 Preproduction sample approval.—Unless otherwise specified (see 6.2), before production is commenced the contractor shall submit for approval to the contracting officer a one-pound sample of feathers, down, or feather and down mixture.

3.3 Materials.—Materials shall be new, not previously used or reclaimed (see 6.2) and shall be well dusted, washed, dried and sterilized.

3.4 Definitions.—The materials covered by this specification are defined as follows:

3.4.1 *Feathers (or natural feathers).*—Bird or fowl plumage having quill shafts and barbs and which has not been processed in any manner other than by washing, dusting. and sterilizing.

3.4.1.1 *Waterfowl feathers.*—Goose feathers, duck feathers or any mixture of goose or duck feathers.

3.4.1.1.1 *Small waterfowl feathers.*—Whole waterfowl feathers, other than quill feathers or damaged feathers, which are less than 2½ inches long.

3.4.1.1.2 *Pin feathers (or nestling feathers).*—A waterfowl feather not fully developed with no distinguishable quill but with relatively short coarse barbs emanating from the sheath.

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3.4.1.1.3 *Waterfowl quill feathers (or quills).*—Wing and tail feathers commonly known and referred to as quills.

3.4.1.1.4 *Plumules.*—Small, downy waterfowl feathers with under-developed quill with barbs or filaments undistinguishable from those of down.

3.4.1.1.5 *Damaged feathers.* — Waterfowl feathers, which are materially broken, materially damaged by insects or otherwise materially injured.

3.4.1.2 *Non-waterfowl feathers.* — The feathers of any kind of fowl other than goose or duck.

3.4.2 *Feather fiber.*—The barbs of feathers which have been completely separated from the quill shaft and any aftershaft, and which are in no wise joined or attached to each other.

3.4.3 *Pure down.* — The undercoating of waterfowl consisting of clusters of the light, fluffy filaments from one quill point but without any quill shaft.

3.4.4 *Down fiber.* — The detached barbs from down plumes or detached barbs from the basal end of feather quill shafts which are undistinguishable from the barbs of down plumes.

3.4.5 *Residual matter.*—Residual matter is defined as quill pith, feather fragments, trash, and foreign matter.

3.4.6 *Percentage composition.*—The percent composition as stated in the applicable paragraph is based on the total weight of the sample tested when tested as specified in table II.

3.5 Requirements applicable to all types.

3.5.1 *Preparation.*—The feathers (type I), down (type II), and feather and down mixture (type III) shall be thoroughly cleaned, washed, dried (see 6.3), and sterilized.

3.5.2 *Blood removal and scouring.*—The finished product shall be entirely free of blood, and a commercial laundry sour of the fluoride type, namely sodium acid or silico fluoride or combinations thereof, shall be utilized in the final rinse (except when DDT treatment is specified as in 3.5.3) of the scouring procedure to bring the acidity (pH) of the finished dry product within the limits of 3.5 to 6.0 when tested as specified in table II.

3.5.2.1 *Certification.*—The contractor shall certify in writing to the use of the fluoride sour in the final rinse of the washing procedures as specified in 3.5.2, and that the materials have been sterilized.

3.5.3 *Insecticide.*—When specified in the applicable procurement document (see 6.2), the feathers, down, and feathers and down mixture shall be treated with DDT (dichloro—diphenyl-trichloroethane). The DDT shall be applied during the preparation of the feathers so that when tendered for delivery the feathers, down, or feather and down mixture shall contain not less than 0.50 nor more than 0.75 percent of DDT based on the air-dried weight of the material when tested as specified in table II. The DDT shall be applied from an aqueous emulsion in accordance with "Direction for Use (Feathers)" of Military Specification MIL–C–3985.

3.5.4 *Cleanliness.*

3.5.4.1 *Oxygen number.*—When tested as specified in table II the feathers, down, and feather and down mixture shall have an oxygen number of not more than 16.

3.5.4.2 *Solvent soluble matter.*—Fats, oils, and waxes shall not be added to the feathers, down, and feather and down mixture. Solvent soluble materials, exclusive of DDT content, shall not exceed 2½ percent when tested as specified in table II.

3.5.5 *Brittleness.*—Feathers shall be sufficiently pliable to permit the ends to be

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brought together in the direction of maximum curvature without any fracture (partial or complete) to the quill or rachis. Not more than four feathers shall fail the test specified in table II.

3.6 Requirements applicable to individual types.

3.6.1 *Type I feathers.*

3.6.1.1 *Composition (see table II).*

(*a*) Whole waterfowl feathers
Minimum 80.0 percent

(*b*) Pure down (and plumules)
Minimum 10.0 percent

(*c*) Balance ____ Maximum 10.0 percent

Composed of any combination of the following:

(1) Non-waterfowl feathers
Maximum 8.0 percent

(2) Waterfowl quill-feathers, feather fibers, or damaged waterfowl feathers
Maximum 5.0 percent

(3) Down fiber__Maximum 1.0 percent

(4) Residual matter
Maximum 2.0 percent

3.6.1.1.1 *Class 1 goose feathers (grade A and grade B).*—A minimum of 50 percent of all feathers shall measure less than 3 inches in length. Feathers longer than 4 inches but not exceeding 6 inches shall total not more than 5 percent. The balance shall be composed of feathers not exceeding 4 inches in length.

3.6.1.1.2 *Class 2 duck feathers (grade A and grade B).*—A minimum of 50 percent of all feathers shall measure less than 2½ inches in length. Feathers longer than 3½ inches but not exceeding 5 inches shall total not more than 5 percent. The balance shall be composed of feathers not exceeding 3½ inches in length.

3.6.1.2 *Filling power (type I).*—The filling power of type I feathers, when measured as specified in 4.4.5, shall be as follows:

Class of feathers	Grade	Filling power (cms)
Class 1—Goose feathers__	A	6.0 (Minimum)
Class 1—Goose feathers__	B	5.0 (Minimum)
Class 2—Duck feathers___	A	5.0 (Minimum)
Class 2—Duck feathers___	B	4.0 (Minimum)

3.6.2 *Type II down.*

3.6.2.1 *Composition (see table II).*

(*a*) Pure down (and plumules)
Minimum 85 percent

(*b*) Balance _____ Maximum 15 percent
Composed of any combination of the following:

(1) Down fiber__Maximum 8.5 percent

(2) Feather fiber
Maximum 5.0 percent

(3) Small waterfowl feathers
Maximum 10.0 percent

(4) Quill feathers, insect damaged feathers and on-waterfowl feathers __Maximum 1.0 percent

(5) Residual matter
Maximum 2.0 percent

3.6.2.2 *Filling power (type II).*—The filling power for type II down, when measured as specified in 4.4.5, shall be as follows:

Class	Grade	Filling power (cms)
1 and 2_____	A	9.0 (Minimum)
1 and 2_____	B	7.5 (Minimum)

3.6.3 *Type III feather and down mixture.*

3.6.3.1 *Composition (see table II).*

(*a*) Pure down (and plumules)
Minimum 40.0 percent

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(*b*) Extraneous materials
Maximum 12.0 percent

Composed of any combination of the following:

(1) Non-waterfowl feathers
Maximum 4.8 percent

(2) Down fiber__Maximum 4.0 percent

(3) Waterfowl quill feathers, feather fibers, insect or otherwise damaged waterfowl feathers
Maximum 5.4 percent

(4) Residual matter
Maximum 2.0 percent

(*c*) Balance to be composed of whole waterfowl feathers.

3.6.3.1.1 A minimum of 50 percent by weight of the feathers (excluding down content) shall measure less than 3 inches. Feathers longer than 4 inches but not exceeding 6 inches shall total not more than 5 percent. The balance of the feathers shall be composed of feathers not exceeding 4 inches in length.

3.6.3.2 *Filling power (type III).*—The filling power of type III feather and down mixture shall not be less than 6.0 cms. when tested as specified in table II.

3.7 Packaging and packing.

3.7.1 *Materials purchased by a contractor.*—When the feathers, down, or feather and down mixture are purchased by a contractor as a component of an end item, they shall be packaged and packed in clean condition in accordance with the feather and down supplier's commercial practice.

3.7.2 *Materials purchased directly by the Government.*—When the feathers, down, or feather and down mixture are purchased directly by the Government, they shall be packaged and packed in the levels specified in accordance with the requirements of section 5 and inspected in accordance with 4.5 through 4.6.1.4.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Contractors inspection responsibility.—Unless otherwise specified herein, the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any other commercial laboratory acceptable to the Government. Inspection records of the examination and tests shall be kept complete and made available to the Government as specified in the contract or order.

4.2 Pre-acceptance inspection.

4.2.1 *Laboratory report.*—The pre-award laboratory report shall contain actual test results showing that the feathers, down, or feather and down mixture have been tested and comply with all physical and chemical requirements of section 3.

4.2.2 *Preproduction sample inspection.*—The preproduction sample submitted in accordance with 3.2 shall be examined and tested to determine compliance with the requirements of section 3.

4.3 Inspection for acceptance. — Unless otherwise indicated, inspection shall be in accordance with Military Standard MIL–STD–105 except as otherwise indicated hereinafter.

4.4 Testing.—Testing of the end item shall be performed in accordance with methods specified in table II. The test methods indicated by number apply to Federal Specification CCC–T–191. The sample unit (unit of product) for testing shall be a 12-ounce composite of feathers, down, or feather and down mixture as applicable. The sample shall be obtained by drawing approximately 4 ounces each of the material from the top, center and bottom on two opposite sides of each bag selected for sampling, so that approximately 24 ounces of material will be drawn from each sample bag. Bags shall be cut in the

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specified locations for drawing the sample. The sample size (number of bags to be selected) shall be in accordance with level L–7, table IIIB of Military Standard MIL–STD–105, and the lot size shall be expressed in terms of bags. For civil agencies the sample size and the number of defects acceptable shall be as specified in table I.

TABLE I.—*Sampling for test — civil agencies*

Number of bags in lot	Number of bags in test sample	Acceptance number (defects)	Rejection number (defects)
25 and under	2	0	1
26 to 65	3	0	1
66 to 110	5	0	1
111 to 300	7	0	1
301 to 500	10	1	2
501 to 800	15	1	2
801 to 1300	25	2	3
1301 to 3200	35	3	4
3201 to 8000	50	4	5
8001 to 22000	75	6	7
22001 and over	110	8	9

TABLE II.

Characteristic	Requirement reference	Test method	Determinations per sample unit
Composition:		(see 4.4.4)	1
Type I	3.6.1		
Type II	3.6.2		
Type III	3.6.3		
Filling power:		(see 4.4.5)	3
Type I	3.6.1.2		
Type II	3.6.2.2		
Type III	3.6.3.2		
DDT content	3.5.3	(see 4.4.3)	2
Acidity (pH)	3.5.2	2811	2
Oxygen number	3.5.4.1	(see 4.4.2.1)	2
Solvent soluble matter	3.5.4.2	(see 4.4.2.2)	2
Brittleness	3.5.5	(see 4.4.1)	1

4.4.1 *Brittleness test.*—Fifty feathers shall be tested to determine compliance with the requirement specified in 3.5.5. The ends of each feather shall be brought together in the direction of maximum curvature. Fracture of the quill or rachis shall be evidence of failure.

4.4.2 *Cleanliness test.*

4.4.2.1 *Oxygen number.* — Ten grams of feathers, down, or feather and down mixture shall be placed in a tumbler jar as specified in method 5500 of Federal Specification CCC–T–191, or in an equivalent agitating device, with one liter of distilled water and tumbled at room temperature for one hour. The resulting suspension shall be filtered through a 74-micron (standard No. 200) sieve conforming to Federal Specification RR–S–366, and evaluated as specified in 4.4.2.1.1.

4.4.2.1.1 A 100-milliliter aliquot of the above filtrate shall be transferred to a porcelain casserole, neutralized and made acid with .1 to 2 milliliters excess of 6*N* sulfuric acid. The solution shall be titrated with standard 0.*IN* potassium permanganate, adding approximately 0.05 milliliters at a time, until a pink color persists for 60 seconds. Calculate oxygen number to number of grams of oxygen per 100,000 grams of sample as follows: Oxygen number = milliliters of 0.*IN* $KMnO_4$ $\times$ 80.

4.4.2.2 *Solvent soluble matter.* — Three grams plus or minus 0.5 gram of feathers, down, or feather and down mixture shall be placed in a fat-free extraction thimble (Whatman single thickness) to prevent escape of sample. Thimble with product shall be extracted in a Soxhlet extractor, using carbon tetrachloride or chloroform for a minimum of 20 cycles. The extract shall then be evaporated on a water bath, dried to constant weight in an oven at 100° ± 2° C, cooled and weighed. Report residue, after deducting DDT as determined in 4.4.3 if present, as the solvent soluble matter.

4.4.3 *DDT (dichloro-diphenyl-trichloroethane) content.* — A 10 ± 1-gram quantity of feathers, down, or feather and down

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mixture shall be extracted for two hours in a Soxhlet apparatus with petroleum ether or any other suitable solvent, after which most of the solvent shall be removed by evaporation on a steam bath. Do not evaporate to dryness, since the DDT may decompose. Add 25 ml. of 99 percent isopropyl alcohol and 2.5 grams of metallic sodium cut into small pieces and swirl the flask in order to mix its contents. Connect to a water-cooled reflux condenser and boil gently for a minimum of one half hour. Shake the flask occasionally. Decompose the excess sodium by cautiously adding 10 milliliters of 50 percent isopropyl alcohol through the condenser at a rate of one to two drops per second. Boil for an additional 10 minutes and then add 60 milliliters of distilled water. Cool to room temperature, add two to three drops of phenolpthalein solution, neutralize by adding nitric acid (1 to 1), and add 10 milliliters of the diluted acid in excess. Add dropwise, with stirring of the solution, a measured excess (25 ml.) of 0.*IN* $AgNO_3$ solution. Coagulate the precipitate by heating on a steam bath for approximately one half hour. Cool to room temperature and filter through a No. 52 Whatman filter paper and wash thoroughly with distilled water, receiving the filtrate in a 500-ml. Erlenmeyer flask. Add 5 milliliters of ferric ammonium sulfate indicator and titrate the excess $AgNO_3$. Compute the net number of milliliters of 0.*IN* $AgNO_3$ consumed by the sample. Calculate percent DDT as follows:

$$\text{Percent DDT (dichloro-diphenyl-trichloroethane)} = \frac{\text{Milliliters of 0.}IN\ AgNO_3\ \text{(consumed)} \times 7.094}{\text{Weight of sample (grams)}}$$

4.4.3.1 A blank determination for the chemicals used shall be made following the exact procedure given above, but limiting the 0.*IN* $AgNO_3$ solution to 5 milliliters in order to obtain a chloride correction value. This blank value shall be subtracted from the DDT determination to obtain the exact value of DDT present.

4.4.4 Composition analysis by weight of feathers, down, and feather and down mixture.

4.4.4.1 *Atmospheric conditions.*—The test for composition analysis may be made under prevailing atmospheric conditions except in the settlement of disputes. In the case of dispute the test shall be performed under standard conditions as specified in Federal Specification CCC–T–191.

4.4.4.2 *Preparation of specimen.*—Specimens of the end product shall be prepared as specified in 4.4. Approximately one ounce of the material shall be exposed in the relaxed state (without packing) in a screen container for a period of not less than one day. The material shall be mixed with a rod over the exposure period to insure complete conditioning.

4.4.4.3 *Apparatus.*

4.4.4.3.1 A wooden box of the following approximate dimensions:

Base—18 inches long by 12 inches wide
Front—6 inches high
Back—12 inches high

The box shall have a glass lid hinged to the top edge of the back panel to provide a clear view of its interior and a free and secure closure during examination of the sample. It shall also have two openings, one at each end of the front panel, sufficiently large to permit the examiner's hand to enter the box to separate and examine the particles in the sample, and shall be equipped with a light to provide sufficient illumination of its interior.

4.4.4.3.2 Tared weighing bottles or beakers with covers to be used as containers in the separating process.

4.4.4.3.3 A pair of tweezers suitable for extracting and separating the components.

4.4.4.3.4 *Balance.*—An analytical balance capable of accurately weighing to the nearest 0.001 gram shall be utilized in the weighings.

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4.4.4.4 *Procedure.*

4.4.4.4.1 The sample consisting of approximately four grams of specimen prepared as in 4.4 and 4.4.4.2 shall be transferred into a weighing container of known weight and shall then be weighed to determine its exact weight.

4.4.4.4.1.1 *Weighing.*—All weights shall be determined to the nearest 0.001 gram.

4.4.4.4.2 The weighing bottle or beaker containing the sample shall be placed on the floor of the separating box along with as many empty containers of known weight as necessary for the separation of the sample. In the event of dispute, prior to their use, all weighing bottles or beakers shall be kept in a conditioned atmosphere as defined in Federal Specification CCC–T–191. The lid of the separating box shall be kept closed.

4.4.4.4.3 Each component of the feathers, down, or feather and down mixture in the sample to be analyzed shall be extracted individually and shall be closely examined to determine its nature. Each particle shall be classified in accordance with the definitions in 3.4 and shall be placed into the appropriate weighed container. The weighing bottles or beakers shall be identified as follows:

(*a*) For type I—waterfowl feathers:
- Whole waterfowl feathers.
- Pure down (and plumules).
- Non-waterfowl feathers.
- Waterfowl quill feathers, feather fibers, or damaged waterfowl feathers.
- Down fiber.
- Residual matter.

(*b*) For type II—down:
- Pure down (and plumules).
- Down fiber.
- Feather fiber.
- Small waterfowl feathers.
- Quill feathers, insect damaged feathers and non-waterfowl feathers.
- Residual matter.

(*c*) For type III—feather and down mixture:
- Pure down (and plumules).
- Whole waterfowl feathers.
- Non-waterfowl feathers.
- Waterfowl quill feathers, feather fiber, insect or otherwise damaged waterfowl feathers.
- Down fiber.
- Residual matter.

4.4.4.5 *Evaluation.*

4.4.4.5.1 Upon completion of the separating process, the weight of the contents of each weighing bottle or beakers shall be determined by subtracting the tare weight of the containers from the gross weight.

4.4.4.5.2 The percentage of each component shall be determined in accordance with the following formula:

$$\frac{\text{Total weight of material in component}}{\text{Weight of sample}} \times 100 = \text{Percent of component sample.}$$

4.4.4.6 *Report.*

4.4.4.6.1 The percent of each component of the mixture shall be reported to the nearest 0.1 percent based on the weight of the sample prior to separation.

4.4.4.6.2 In the event that the sum of the weights of all components varies more than one percent from the original sample weight, the test shall be repeated and the original results discarded.

4.4.4.7 *Preservation of specimens.*

4.4.4.7.1 When test reports indicate failure, each separate portion of the sample shall be kept in a properly identified envelope for ready reference in the event of a dispute. If test results are not questioned within three months from date of the report, these samples may be discarded. If results indicate

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failure, a recheck shall be performed on another specimen obtained in a similar manner from the original sample.

4.4.5 *Determination of filling power.*

4.4.5.1 *Atmospheric conditions.*—The filling power test may be conducted under prevailing atmospheric conditions except in the settlement of disputes. In the case of dispute the test shall be performed under standard conditions as specified in Federal Specification CCC–T–191.

4.4.5.2 *Preparation of specimen.*—Specimens of the end product shall be prepared as specified in 4.4. Approximately three ounces of material shall be fluffed in the filling power apparatus for 2 minutes at 50 ± 10 pounds air-gage pressure. The material shall then be exposed in the unpacked state in a screen container for not less than 3 days nor more than 5 days. The material shall be mixed with a rod once a day to insure complete relaxation and conditioning.

4.4.5.3 *Test apparatus.* — The apparatus (see figure 1) shall consist of the following:

4.4.5.3.1 *Cylinder.*—The cylinder shall be a rigid phenolic bonded paper or aluminum tube 12.75 inches ± 0.01 inch inside diameter, and approximately 20 inches in length. The inner wall shall have a smooth finish. The cylinder shall be set in a vertical position and shall be open at both ends with means of supporting the base end rigidly in a horizontal plane. It is convenient to have the cylinder detachable from the base for facilitating specimen removal. A removable perforated cover with the air-blowing unit attached thereto shall be provided for use in retaining the specimen during air fluffing.

4.4.5.3.2 *Piston.*—The piston shall be made of a rigid material and shall weigh 118.0 grams ± 0.5 gram (Balsa wood with a stiff, smooth paper contact surface or expanded rubber has been found satisfactory for this purpose). The piston shall have a diameter of 12.55 inches ± 0.01 inch with the edge beveled to ⅛ inch. The exact center of the piston shall be well defined for measuring purposes. A device for lowering the piston in a horizontal position shall be provided. This may be accomplished by suitably attached strings or other manual or mechanical means.

4.4.5.3.3 *Measuring device.*—The measuring device shall consist of a smooth thread, with small plumbs as counterweights, one of the plumbs serving in connection with the depth indicator. The thread shall run over two pulleys so that one plumb will drop directly on the center of the piston while the other will run along a vertically fixed centimeter scale.

4.4.5.3.4 *Fluffing device.*—Means for thoroughly fluffing or loosening the specimen shall be provided. A blower is recommended for this purpose.

4.4.5.3.5 *Balance.*—An analytical balance capable of weighing accurately to the nearest 0.001 ounce shall be utilized for specimen weighings.

4.4.5.4 *Procedure.*

4.4.5.4.1 *Determination of zero reading.*—Slowly lower the piston until it rests on the botton of the empty cylinder. Lower the plumb until it just touches the top surface of the piston. The depth is read directly from the centimeter scale to the nearest millimeter. This shall be repeated until two consecutive identical readings are obtained. This measurement shall be read to the nearest 0.1 cm and shall be considered as zero depth (depth of empty cylinder).

4.4.5.4.2 *Test specimen.* — The specimen shall consist of 0.80 ± 0.002 ounce of material prepared as specified in 4.4.5.3.

4.4.5.4.3 *Fluffing.*—This specimen shall be placed in the cylinder and the piston slowly

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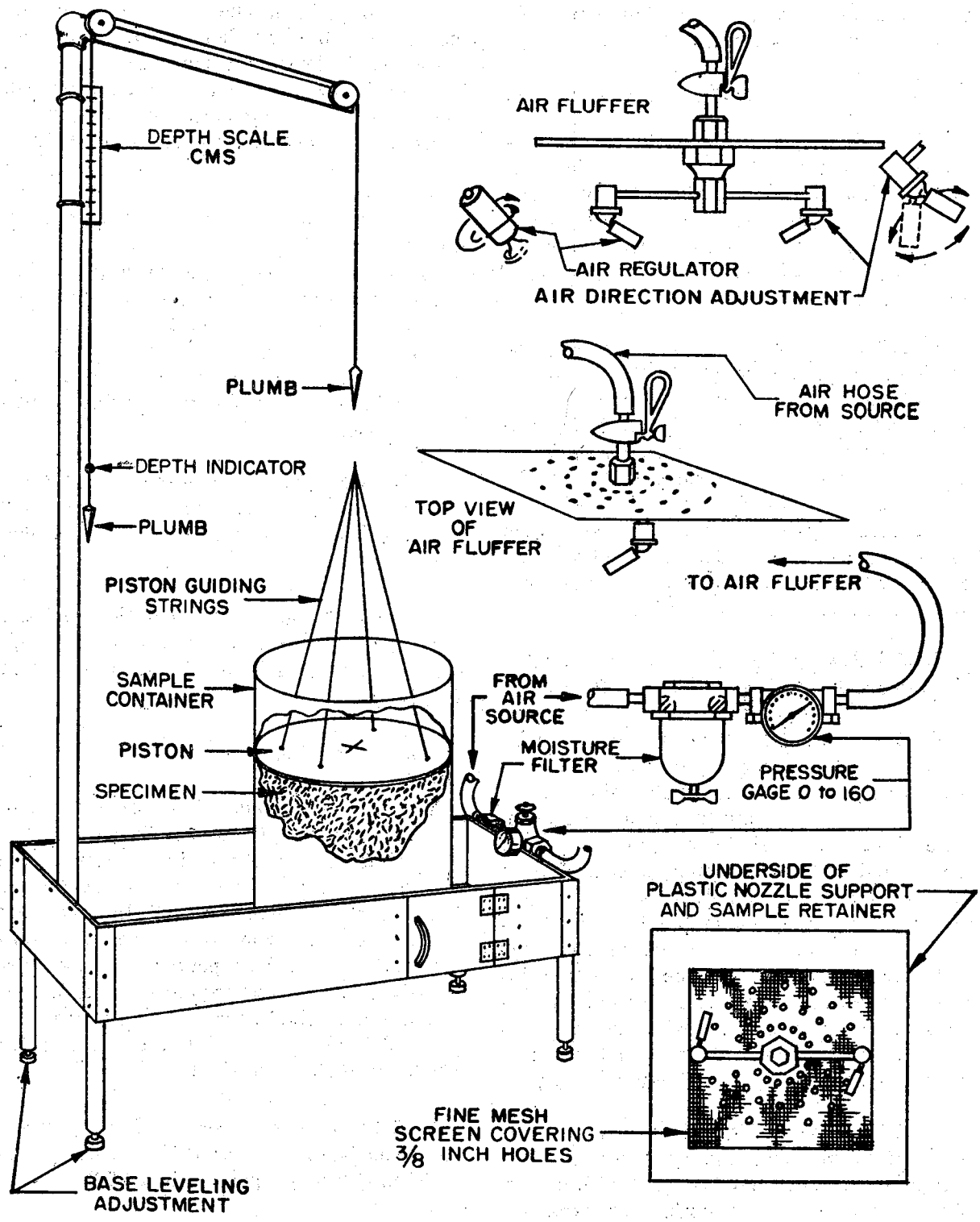


FIGURE 1.

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lowered onto the material, and air blown for approximately 10 seconds at 50 ± 10 pounds air-pressure to thoroughly mix and fluff the specimen. Other means may be used provided results equal to the blower method are obtained.

4.4.5.4.4 *Volume measuring.*—The piston shall be slowly lowered onto the material. When there is a definite slackening of the supporting strings, a stopwatch shall be started. The plumb shall be lowered so that at the end of a 1-minute interval, it just touches the center of the top surface of the piston. The depth shall be read directly from the centimeter scale to the nearest millimeter. Each specimen shall be refluffed and measured three times for depth.

4.4.5.5 *Report.*—The report shall be based on the volume at the designated time pressure level. The distance between inside surface of the bottom of the cylinder and the bottom surface of the piston shall be reported. This shall be obtained by subtracting the zero reading obtained in 4.4.5.4.1 from the depth obtained in 4.4.5.4.4. Unless otherwise specified, three specimens shall be tested from each unit of product. A measure of the height of the material under the piston shall be considered its filling power. The filling power of the unit of product shall be the average of the results obtained from the specimens tested and shall be reported to the nearest 0.1 cm.

4.5 Inspection of components.

4.5.1 *Inspection of cloth and fiberboard.*—Cloth and fiberboard used in the fabrication of fabricated bags and bale boards shall be inspected to determine compliance with the requirements of referenced specifications and applicable supplemental provisions specified in 5.1 and 5.2.

4.5.2 *Inspection of fabricated bags and bale boards.* — Fabricated bags and bale boards shall be inspected to determine conformance with the applicable requirements of 5.1 and 5.2. The acceptable quality level shall be 4.0 defects per hundred units, and the inspection level shall be L–3 of table IIIB of Military Standard MIL–STD–105. For civil agencies the sample size and the number of defects acceptable shall be as specified in table III.

TABLE III.—*Sampling for inspection of fabricated bags and bale boards—civil agencies*

Number of bags or bale boards in lot	Number of bags or bale boards in sample	Acceptance number (defects)	Rejection number (defects)
300 and under	1	0	1
301 to 500	2	0	1
501 to 800	3	0	1
801 to 1300	5	0	1
1301 to 3200	7	0	1
3201 to 8000	10	1	2
8001 to 22000	15	1	2
22001 and over	25	2	3

4.6 Inspection of the end product.

4.6.1 *Examination of the end item.*—The end item (completely packaged and packed product) shall be inspected in accordance with the examinations and tests specified in 4.6.1.1, 4.6.1.2, 4.6.1.3, 4.6.1.4.

4.6.1.1 *Weight.*—The lot shall be unacceptable if the total net weight of the bags in the sample is less than the total net weight indicated on the bags.

4.6.1.2 The sample size (number of bags to be selected) for the weight examination shall be in accordance with level L–8, table IIIB of Military Standard MIL–STD–105. For civil agencies the sample size shall be as specified in table IV. The net weight of the material contained in the sample bags shall be obtained by deducting the tare weights from the gross weights of the sample bags.

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TABLE IV.—*Sampling for weight examination—civil agencies*

Number of bags in lot	Number of bags in sample
25 and under	2
26 to 40	3
41 to 65	5
66 to 110	7
111 to 300	10
301 to 500	15
501 to 800	25
801 to 1300	35
1301 to 3200	50
3201 to 8000	75
8001 to 22000	110
22001 and over	150

4.6.1.3 *Packaging examination.*—The unit of product shall be one bag of feathers, down, or feather and down mixture.

Defects

Not packaged in accordance with specified packaging level.
Bag not new, or any patch.
Not closed in the specified manner.
Wire tie broken or not tightly turned to accomplish a secure closure.
Any hole, rip, tear or open seam; any weak places in bag that may develop into a hole.
Bag wet.
Marking — missing, incomplete, illegible, incorrect, not accomplished in the specified manner, or not in proper location.

4.6.1.4 *Packing examination.*—The sample unit shall be one bale or shipping container.

Defects

Not packed in accordance with the specified packing level.
Bags not compressed to specified size.
Does not contain the specified number of bags.
Kraft facing of baling board not on outside.
Any board or strap missing.
Baling board or strap too loose or not placed in proper location to a degree where safe delivery of contents may be impaired.
Marking — missing, incomplete, illegible, incorrect, not accomplished in the specified manner or not in proper location.

4.6.1.5 Acceptable quality levels.—Acceptable quality levels (AQL's) and inspection levels shall be as follows: AQL's are expressed in defects per 100 units.

Examination	*Inspection level*	*AQL*
4.6.1.3 (packaging)	I	6.5
4.6.1.4 (packing	1	6.5

For civil agencies the sample size and the number of defects acceptable shall be as specified in table V.

TABLE V.—*Sampling for packaging and packing examination—civil agencies*

Number of bags or bales in lot	Number of bags or bales in sample	Acceptance number (defects)	Rejection number (defects)
110 and under	5	0	1
111 to 300	7	1	2
301 to 500	10	1	2
501 to 800	15	2	3
801 to 1300	25	3	4
1301 to 3200	35	5	6
3201 to 8000	50	6	7
8001 to 22000	75	9	10
22001 and over	110	12	13

5. PREPARATION FOR DELIVERY

Federal Standard No. 102 should be referred to for definitions and applications of the various levels of packaging and packing protection for supplies and equipment (for civil agencies only).

5.1 Packaging.—Packaging shall be level A or C as specified (see 6.2).

5.1.1 *Level A.*—Fifty pounds ± 5 pounds of type I feathers of one class and grade only, or 50 pounds ± 5 pounds of type III feathers and down mixture, or 25 pounds ± 3 pounds of type II down of one class and grade only shall be packaged in a new (not used or laundered) plain textile bag fabricated from cotton sheeting conforming to type I, class

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D of Military Specification JAN–C–485 or cloth, cotton, osnaburg conforming to type I, class 2, finish A of Federal Specification CCC–C–429. Fabrics in other widths of equivalent weight and construction will be acceptable. For civil agencies only, the package shall be 25 pounds of down (type II) and 50 pounds of feathers (type I) with a minus tolerance of 4 ounces.

5.1.1.1 *Bag construction.*—The bags shall have a finished length and width measured flat of approximately 72 by 36 inches and shall be constructed in accordance with the following requirements: The side and bottom shall be seamed with 5 to 6 stitches per inch with either seam and stitching type 401–SSa–2 or 301–SSa–2, conforming to Federal Standard No. 751. The stitching shall be at least ⅝ inch from the cut edge of the fabric or ¼ inch from a selvage edge and shall be at least ⅛ inch apart. The ends of all seams shall be backstitched at least 1 inch. The thread used for seaming shall be cotton, machine, soft finish, type IIIA, unbleached, ticket no. 10/3, 3 ply, conforming to Federal Specification V–T–276 (table V).

5.1.1.1.1 *Type 301 stitching.* — Thread breaks in stitching shall be backstitched not less than 1 inch at each break. Thread tension shall be maintained so that there will be no loose stitching and that the lock will be embedded in the center of the materials sewed. Thread ends shall be trimmed.

5.1.1.1.2 *Type 401 stitching.* — Thread breaks in stitching shall be overstitched not less than 1½ inches at each break. Thread tension shall be maintained so that there will be no loose stitching. The chain portion of the stitching shall not appear on the outside of the bag. Thread ends shall be trimmed.

5.1.1.2 *Closure.*—The closure of the top of the bag after filling shall be effected by means of two 6-inch, 15-gage, soft-iron or steel-wire ties having a ½-inch-diameter formed eye at each end. One tie shall be applied at the base of an ear formed by gathering the cloth evenly and the second tie applied at a distance approximately 1 inch from the first tie with the twisted ends of two ties alternated and facing in opposite directions.

5.1.2 *Level C.*—Feathers, down, and feathers and down mixed shall be packaged in accordance with good commercial practice.

5.2 Packing.—Packing shall be levels A, B, or C as specified (see 6.2).

5.2.1 *Levels A and B.*—Three 50-pound bags of type I feathers of one class and grade only, or three 50-pound bags of type III feathers and down mixed or four 25-pound bags of type II down of one class and grade only, packaged, with permissible tolerances as specified in 5.1.1, shall be placed between two baling boards and compressed by machine so as to form a bale approximately one-fifth of the original volume. Bales thus formed shall measure approximately 60 by 24 by 20 inches.

5.2.1.1 *Baling boards.*—Baling boards shall be not less than 0.075 inch thick solid fiberboard conforming to type II, class 1, of Federal Specification PPP–B–636 except that at least one face of the board shall be not less than 0.010-inch kraft paperboard, and the bursting strength of the combined board shall be not less than 200 p.s.i. Two such baling boards size 58 by 36 inches, each with a score line placed 6 inches from each 58-inch edge in such a manner that the kraft facing is on the outside when in a folded position, shall be used in each bale; one board on top of the bags being baled and one on the bottom. The boards shall be bent at the score lines thereby forming a fiberboard cap 58 inches in length, 24 inches in width, with 6-inch flanges extended down over the side of the bale.

5.2.1.2 *Strapping.* — Not less than eight straps shall be used on each bale; straps shall be affixed girthwise around the sides, top, and bottom of the bale over the baling boards. The two outside straps shall be placed 4 inches in from each end and the intermediate straps placed equidistant between the end bands. All strapping shall be flat steel not less

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than 3/8 inch wide by 0.020 inches thick conforming to type I, class A, of Federal Specification QQ–S–781.

5.2.2 *Level C.*—Feathers, down, and feathers and down mixture packaged as specified in 5.1 shall be packed in a manner to insure carrier acceptance and safe delivery at destination. Containers shall be in accordance with Uniform Freight Classification Rules or Regulations of other carriers applicable to the mode of transportation.

5.3 Marking.

5.3.1 *Civil agencies.*—In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with the requirements of Federal Standard No. 123.

5.3.2 *Military agencies.* — In addition to any special marking required by the contract or order, shipments shall be marked in accordance with the requirements of Military Standard MIL–STD–129. All shipments for level A or level B shall be designated "A/A".

6. NOTES

6.1 Intended use.—The type I feathers and type II down are intended for storage for subsequent use in end items as required. The type III feather and down mixture covered by this specification is intended for immediate use in sleeping bags and similar items.

6.2 Ordering data.—Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title, number and date of this specification.
(*b*) Type, class, and grade required (see 1.2
(*c*) When laboratory report is not required (see 3.1).
(*d*) When preproduction sample is not required (see 3.2).
(*e*) Whether an affidavit is required to show that the materials supplied are new and not used (see 3.3).
(*f*) When treatment with insecticide is required (see 3.5.3).
(*g*) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 With reference to 3.5.1, good results have been obtained when the feathers or down are agitated for 20 minutes in water at 90° ± 5° F. to which has been added a blood solubilizing agent in the proportion of one ounce of the agent to 10 gallons of water. Following the blood solubilizing treatment, the feathers or down are washed using a non-ionic detergent and then rinsed. The final rinse contains one of the commercial fluoride sours available as laundry sours, namely, sodium acid fluoride or sodium silica fluoride or combinations thereof.

6.4 Supersession data.—This specification supersedes those requirements of Federal Specification C–F–151a dated July 24, 1940 pertaining to type II duck feathers and type III goose feathers. This specification also supersedes the requirements of Military Specification MIL–F–5652C dated 29 May 1953 for Feathers and Down, Waterfowl. Federal Specification C–F–158 for Feathers, Land Fowl (Crushed) supersedes those requirements of Federal Specification C–F–151a for type I chicken feathers.

6.5 Filling power test.—The testing equipment may be procured at:

CUSTOM SCIENTIFIC INSTRUMENT, INCORPORATED
541 Devon Street
Arlington, New Jersey

Other sources of equipment may be used provided that the equipment conforms to the standard design or that test results equal to the standard are obtained. Manufacturers are invited to advise the Military Clothing and Textile Supply Agency, Philadelphia Quartermaster Depot, of their readiness to supply

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such testing equipment in order that prospective bidders under this specification may be advised upon their request.

6.6 Transportation description.—Transportation description applicable to this item is:

Feathers NOIBN (specify machine or not machine pressed).

Carload minimum weight 15,000 pounds, subject to rule 34,

Uniform Freight Classification.

Less than 4 lbs. per cubic foot.

Motor volume minimum weight, any quantity.

4 lbs. per cubic foot but less than 7 lbs.

Truckload minimum weight 5,600 pounds, subject to rule 34, National Motor Freight Classification.

7 lbs. per cubic foot but less than 12 lbs.

Truckload minimum weight 10,000 pounds, subject to rule 34, National Motor Freight Classification.

12 lbs. per cubic foot or over.

Truckload minimum weight 12,900 pounds, subject to rule 34, National Motor Freight Classification.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded, by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTEREST:
Army—MQ
Navy—S Md
Air Force.

Copies of this specification may be purchased for cents each.

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